

# APAC Luxury Retail Performance Model: Technical Specification

Sep 23, 2026 · @Jerome

## 1. Overview

This Power BI model answers one executive question: why did APAC net sales fall 14% in 2025 (USD 4.10M to 3.51M) after four years of growth, and which levers recover it. The source is a synthetic star schema for a fictional luxury house: 8 markets, 16 boutiques plus e-commerce, Jan 2021 to Dec 2025.

### Audience

| Persona                                | Decision they make                                       | Primary page                   |
|----------------------------------------|----------------------------------------------------------|--------------------------------|
| Regional MD / Commercial Director      | Where to intervene across markets and divisions          | Executive Summary              |
| Retail Excellence / Boutique Directors | Which stores and advisors to coach                       | Store and Advisor Productivity |
| CRM / Clienteling lead                 | Which client cohorts to reactivate and protect           | Client and CRM                 |
| Merchandising                          | Which categories, lines and SKUs to push, deepen or exit | Category and Product           |
| Supply / Allocation                    | Where stock sits against demand                          | Inventory (extension table)    |

### Business questions in scope

1. Performance: net sales, growth, margin by market, store, channel, division, with 2025 variance decomposed into volume, price and mix.
2. Client: retention, reactivation, cohort value, VIC concentration, tier migration, tourist versus domestic dependency.
3. Productivity: sales per advisor, clienteling reach, book coverage, attribution quality.
4. Category: division and line mix, entry product conversion into higher divisions, price band and markdown exposure.
5. Inventory: stock cover, sell through, aged stock (requires the extension table in Section 4).

**Out of scope:** forecasting beyond a simple trend, footfall conversion (no traffic data), real FX impact (rates are static per store).

## 2. Source data

One workbook (luxury\_retail\_star\_schema.xlsx), six data sheets plus README. One fact table, five dimensions, all foreign keys resolve with zero orphans.

| Table       | Type                   | Grain                             | Rows   | Key                 | Notes                                                                                |
|-------------|------------------------|-----------------------------------|--------|---------------------|--------------------------------------------------------------------------------------|
| fact_sales  | Fact                   | One transaction line              | 14,211 | sales_line_id       | 5 FKs; returns are negative qty and value; 437 return lines                          |
| dim_client  | Dimension              | One client                        | 2,001  | client_key          | client_key = -1 is Anonymous; 897 Prospects never bought                             |
| dim_product | Dimension              | One SKU (model, colour, material) | 280    | product_key         | 5 divisions, 8 categories, 3 price bands; 77 permanent SKUs have no end_of_life_date |
| dim_staff   | Dimension (SCD2 ready) | One staff version                 | 113    | staff_key           | 96 people, 16 shared boutique logs, 1 e-commerce placeholder                         |
| dim_store   | Dimension              | One store                         | 17     | store_key           | 16 boutiques (8 flagship, 8 boutique) plus ECOM; static fx_to_usd                    |
| dim_date    | Dimension              | One day                           | 1,826  | date_key (YYYYMMDD) | 2021-01-01 to 2025-12-31                                                             |

### fact\_sales columns by role

| Role                  | Columns                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| Keys                  | sales_line_id, transaction_number, line_number, date_key, client_key, product_key, store_key, staff_key |
| Degenerate attributes | channel (Boutique, Remote Sale, E-commerce), transaction_type (Sale, Return), currency                  |
| Local measures        | quantity, gross_sales_local, discount_local, net_sales_local                                            |
| USD measures          | fx_to_usd, net_sales_usd, cogs_usd                                                                      |

| Role  | Columns                                                                   |
|-------|---------------------------------------------------------------------------|
| Flags | is_tax_free, is_markdown, is_gift,<br>is_employee_purchase, is_new_client |

Volume by year

| Year | Net sales USD | Transactions | Lines |
|------|---------------|--------------|-------|
| 2025 | 3,509,704     | 1,974        | 3,333 |
| 2024 | 4,101,157     | 1,912        | 3,220 |
| 2023 | 3,466,585     | 1,764        | 2,915 |
| 2022 | 2,931,919     | 1,618        | 2,687 |
| 2021 | 2,329,997     | 1,232        | 2,056 |

Note the 2025 pattern: transactions up 3% while net sales fell 14%. The decline is basket value and mix, not traffic. This is the analytical spine of the case study.

3. Data quality findings and handling rules

Referential integrity is clean and USD conversion reconciles to within 0.005. The issues below are business realities, each with an explicit handling rule and a visible KPI so users can see the quality, not just trust it.

| #   | Finding                                                   | Scale                           | Handling rule                                                                             | Surfaced as             |
|-----|-----------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------|-------------------------|
| DQ1 | Anonymous sales<br>(client_key = -1)                      | 12.7% of lines,<br>13.1% of USD | Keep in sales totals; exclude from all client counts, retention, cohort and tier measures | Client Capture Rate KPI |
| DQ2 | Shared POS logins<br>(Boutique Team) and ECOM placeholder | 17 keys, 8.6% of USD            | Exclude from advisor metrics (role_type is not Placeholder); include in store totals      | Attribution Rate        |

| #   | Finding                                            | Scale                                                                                                                                                      | Handling rule                                                                    | Surfaced as                                          |
|-----|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------|
| DQ3 | Clients assigned to resigned advisors              | 70 clients, 4 resigned advisors; 19 still bought in 2025 (\$82.5K)                                                                                         | Orphaned book; reassignment list                                                 | Orphaned Clients card; Active Orphaned Clients table |
| DQ4 | Missing birth_year                                 | 399 clients                                                                                                                                                | age_band = Unknown, kept as a visible category                                   | None                                                 |
| DQ5 | Missing email / phone                              | 210 / 328 clients                                                                                                                                          | Contactable = channel opt in AND matching contact present (messaging uses phone) | Contactable %                                        |
| DQ6 | Stored client_tier vs recomputed 2025 net spend    | 14 mismatches (11 tiered clients with 2025 net spend at or below zero after returns, recomputed as Lapsed; 3 boundary cases: 2 VIC to Core, 1 VVIC to VIC) | Recompute tier in DAX; keep stored tier for comparison                           | QA test 5                                            |
| DQ7 | Tourists recorded as residence = Overseas          | 24% of clients flagged tourist                                                                                                                             | is_tax_free for transaction level tourist sales; is_tourist for client level     | Tourist Share                                        |
| DQ8 | Static FX per store                                | 1 rate per currency                                                                                                                                        | All USD treated as constant currency                                             | Footer note                                          |
| DQ9 | dim_staff SCD2 columns present but no history rows | 113 rows, all current                                                                                                                                      | Modelled as Type 1; SCD2 columns retained                                        | None                                                 |

| #    | Finding                                             | Scale                                         | Handling rule                                                                                                                                                   | Surfaced as |
|------|-----------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| DQ10 | Returns not linked to original sale                 | 437 lines, 3.1% of gross USD                  | Netted in the period they occur; separate Return Rate. Returns on anonymous tickets can push identified share above 100%, so capture ratios use sale lines only | Return Rate |
| DQ11 | is_employee_purchase always FALSE                   | 0 rows                                        | Column retained, no visual                                                                                                                                      | None        |
| DQ12 | Permanent SKUs with null end_of_life_date           | 77 SKUs                                       | Keep null; lifecycle_status = Permanent; date logic excludes blanks                                                                                             | None        |
| DQ13 | season_year holds the text 'N/A' for permanent SKUs | 77 SKUs                                       | Replace 'N/A' with null before the Whole Number type step (otherwise load errors)                                                                               | Clean load  |
| DQ14 | Dates stored as Excel serial numbers                | dim_client, dim_staff, dim_store date columns | Converted from Whole Number to Date in Power Query                                                                                                              | None        |
| DQ15 | fact_sales cogs_usd loaded as text                  | All rows                                      | Set to Fixed Decimal; before the fix COGS and GM % failed                                                                                                       | None        |

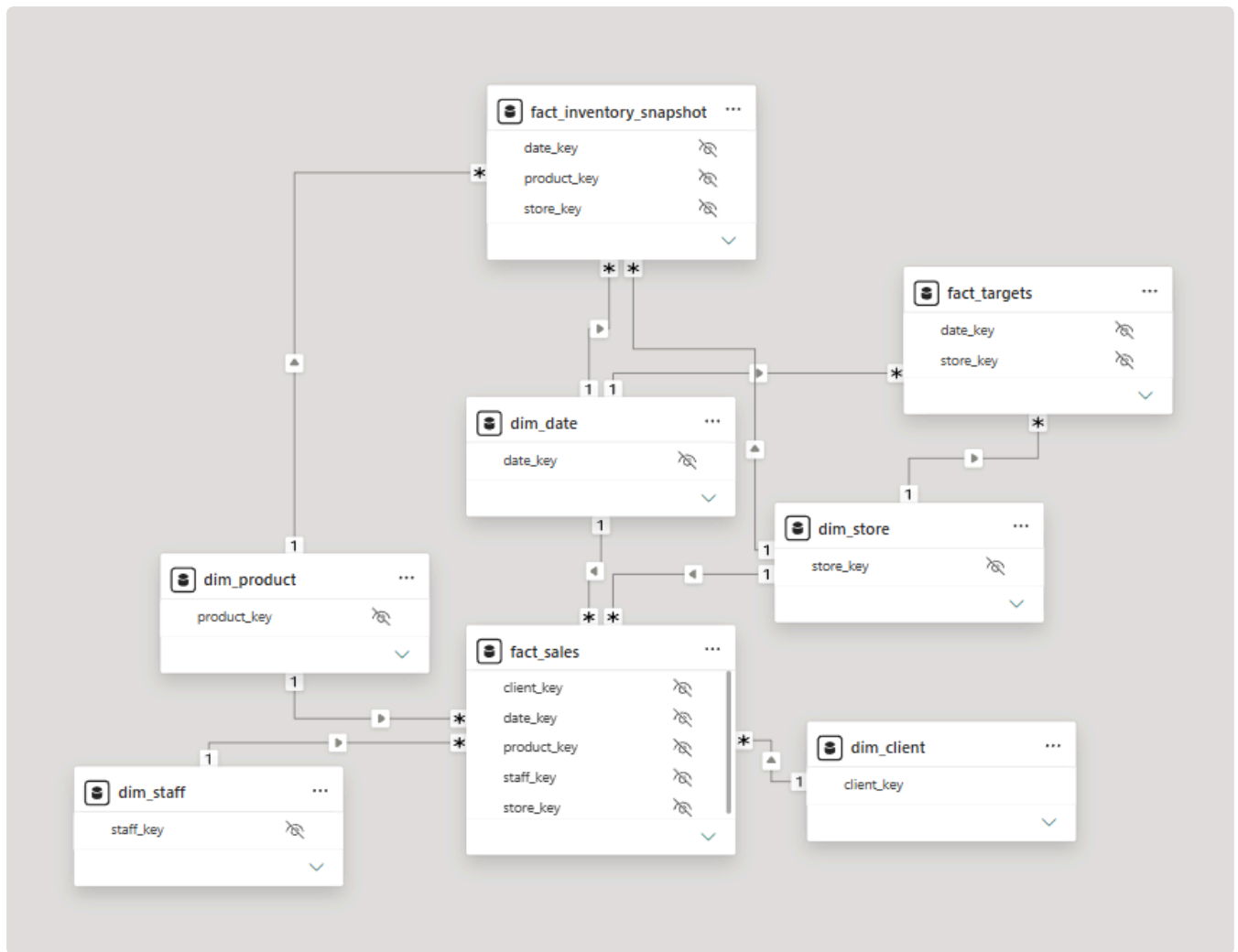
All counts were profiled in Python against the source workbook on the as-of date. Re-run the profile script (repo /sql or /python) after any data refresh.

## 4. Semantic model

Star schema, single direction filters, one active path per dimension. Two extension facts (inventory, targets) turn it into a fact constellation sharing conformed dimensions.

**Model inventory (as built in LinkedIn\_Case\_Study.pbix)**

| Object                                                              | Type                            | Source                                                            | Rows / items         |
|---------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------|----------------------|
| fact_sales, dim_client, dim_product, dim_staff, dim_store, dim_date | Import tables                   | luxury_retail_star_schema.xlsx                                    | As Section 2         |
| fact_inventory_snapshot                                             | Import table (extension)        | luxury_retail_extensions.xlsx, generated by generate_inventory.py | 108,964 rows         |
| fact_targets                                                        | Power Query table (extension)   | Referenced from fact_sales query                                  | 798 rows             |
| Time Intelligence                                                   | Calculation group               | Model                                                             | 7 items              |
| Metric Selector, Dimension Selector                                 | Field parameters                | DAX                                                               | 5 options each       |
| Tier Band                                                           | Calculated table (disconnected) | DATATABLE                                                         | 4 rows               |
| _Sales, _Client, _Productivity, _Category, _Inventory, _QA          | Measure tables                  | Enter Data / DAX                                                  | 66 measures in total |
| Tier 2025 Recomputed                                                | Calculated column on dim_client | DAX                                                               | 1 column             |



## Relationships

| From (many)                                 | To (one)                 | Active | Status            | Purpose                     |
|---------------------------------------------|--------------------------|--------|-------------------|-----------------------------|
| fact_sales[date_key]                        | dim_date[date_key]       | Yes    | Built             | Time intelligence           |
| fact_sales[client_key]                      | dim_client[client_key]   | Yes    | Built             | Client attributes           |
| fact_sales[product_key]                     | dim_product[product_key] | Yes    | Built             | Category analysis           |
| fact_sales[store_key]                       | dim_store[store_key]     | Yes    | Built             | Store and market            |
| fact_sales[staff_key]                       | dim_staff[staff_key]     | Yes    | Built             | Selling advisor             |
| fact_inventory_snapshot[date_key]           | dim_date[date_key]       | Yes    | Built             | Stock position at month end |
| fact_inventory_snapshot[product_key]        | dim_product[product_key] | Yes    | Built             | Stock by SKU and category   |
| fact_inventory_snapshot[store_key]          | dim_store[store_key]     | Yes    | Built             | Stock by store              |
| fact_targets[date_key] (first day of month) | dim_date[date_key]       | Yes    | Built             | Budget vs actual by month   |
| fact_targets[store_key]                     | dim_store[store_key]     | Yes    | Built             | Budget vs actual by store   |
| dim_staff[home_store_key]                   | dim_store[store_key]     | No     | Virtual (TREATAS) | Headcount by store          |
| dim_client[preferred_store_key]             | dim_store[store_key]     | No     | Virtual (TREATAS) | Client book by store        |

| From (many)                      | To (one)             | Active | Status            | Purpose                              |
|----------------------------------|----------------------|--------|-------------------|--------------------------------------|
| dim_client[assigned_advisor_key] | dim_staff[staff_key] | No     | Virtual (TREATAS) | Advisor book coverage, orphaned book |

The model holds 10 physical relationships, all many to one and single direction. The three virtual relationships are not created in the model; measures apply them with TREATAS (Section 6.5).

## Power Query transformations

Transformations are minimal by design. Flags that only feed counts are measures (Section 6); a column exists only where a slicer, axis or sort order needs it.

1. fact\_sales: explicit types on every column (keys and line\_number Whole Number, sales\_line\_id and transaction\_number Text, amounts Fixed Decimal); cogs\_usd set to Fixed Decimal (DQ15); fx\_to\_usd removed (held in dim\_store).
2. dim\_client: serial date columns converted to Date; cohort\_year added with `Date.Year([first_purchase_date])` as Whole Number; advisor\_notes removed; opt in columns typed True/False.
3. dim\_product: blank rows removed; season\_year 'N/A' replaced with null before typing (DQ13); end\_of\_life\_date to Date (null kept for permanent SKUs); retail and cost to Fixed Decimal; lifecycle\_status added against AsOfDate (77 Permanent, 192 Discontinued, 11 Active); price\_band\_sort added (Entry 1, Core 2, Premium 3) and set as Sort by Column for price\_band.
4. dim\_date: date converted to Date and marked as the date table; month\_name sorted by month\_num; Calendar hierarchy (year, quarter, month\_name, date); month\_num and week\_of\_year hidden; Auto date/time turned off for the file.
5. dim\_staff and dim\_store: serial dates converted; blank trailing rows filtered out.
6. fact\_inventory\_snapshot: loaded from luxury\_retail\_extensions.xlsx; snapshot\_date to Date, cost and retail to Fixed Decimal.
7. fact\_targets: built from a reference to fact\_sales (Extension tables below).
8. Parameters: AsOfDate (Date, 2025-12-31), SourcePath and ExtensionPath (Text, workbook paths used in every Source step).

## Open items from the model review

| #  | Item                                      | Status | Resolution                              |
|----|-------------------------------------------|--------|-----------------------------------------|
| O1 | Auto date/time created hidden date tables | Closed | Turned off; 0 hidden date tables remain |

| #   | Item                                                                        | Status      | Resolution                                                                                                                    |
|-----|-----------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| O2  | Source paths hard coded to a local user folder                              | In progress | SourcePath and ExtensionPath created; apply in all 7 Source steps and set a neutral path before publishing to GitHub          |
| O3  | fact_sales keys and IDs loaded as text                                      | Closed      | Explicit types set                                                                                                            |
| O4  | cohort_year and lifecycle_status untyped                                    | Closed      | Whole Number and Text; column renamed to lifecycle_status                                                                     |
| O5  | Measures stored in fact_sales                                               | Open        | Only Selector Title remains; move to _Sales                                                                                   |
| O6  | Placeholder column named Column in DAX measure tables (including Headlines) | Open        | Hide it so each shows as a measure table                                                                                      |
| O7  | Day of Week Num                                                             | Not needed  | No weekday visual in v1                                                                                                       |
| O8  | season_year load errors                                                     | Open        | Replace 'N/A' with null before Changed Type; delete the Query Errors helper query                                             |
| O9  | Copied role_type filter on the Inventory heatmap and combo chart            | Open        | Remove it: it drops shared login units (3,421 vs 3,661 in 2025) and inflates Weeks of Cover (heatmap total 32.6 vs card 32.3) |
| O10 | QA page presentation                                                        | Open        | Hide the page; format Expected and QA Actual as #,##0.### so test 11 shows 0.777                                              |

### Extension tables (to be generated, synthetic)

| Table                   | Grain                 | Columns                                                                                                                                                                    | Built with                                                                                   |
|-------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| fact_inventory_snapshot | Month end, store, SKU | date_key, snapshot_date, store_key, product_key, opening_units, units_received, units_sold_net, units_transferred_out, on_hand_units, on_hand_cost_usd, on_hand_retail_usd | Python generator (generate_inventory.py, seed 42), loaded from luxury_retail_extensions.xlsx |

| Table              | Grain        | Columns                                                                   | Built with                                                                                                                                                                                  |
|--------------------|--------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fact_targets       | Month, store | date_key (first of month), store_key, sales_target_usd, new_client_target | Power Query, referencing fact_sales: prior year actual by store and month times budget growth (2022 +20%, 2023 +15%, 2024 +12%, 2025 +10%), negative months floored at zero, rounded to 100 |
| Measure tables     | n/a          | _Sales, _Client, _Productivity, _Category, _Inventory, _QA                | Enter Data (_Sales) and DAX placeholder tables                                                                                                                                              |
| Time Intelligence  | n/a          | Period, Ordinal; items CY, PY, YoY, YoY %, YTD, PYTD, R12M                | Calculation group (Section 6.2)                                                                                                                                                             |
| Metric Selector    | n/a          | Net Sales, Units Sold, Transactions, Active Clients, GM %                 | Field parameter (Section 6.9)                                                                                                                                                               |
| Dimension Selector | n/a          | Market, Store, Division, Category, Channel                                | Field parameter (Section 6.9)                                                                                                                                                               |
| Tier Band          | 4 rows       | Tier, Min USD, Max USD, Sort                                              | Disconnected DATATABLE (Section 6.4)                                                                                                                                                        |

Inventory generation logic: each month, closing = opening + units received minus net units sold minus units transferred out. Receipts are a launch allocation then monthly replenishment to a target cover of 1.0 to 2.5 months per store, never below the units needed to cover that month's sales; leftover stock is transferred out 3 months after end of life. Net units sold are taken from fact\_sales, so the table reconciles exactly (15,150 units) and never goes negative. Dec 2025: 2,476 units on hand, about 35 weeks of cover, 60% sell through, 7.9% aged stock by cost.

## 5. Business definitions

Every KPI below has one owner definition. Measures in Section 6 implement these exactly; any change here must change the DAX.

| KPI            | Definition                                | Rule / edge case              |
|----------------|-------------------------------------------|-------------------------------|
| Net Sales USD  | Sum of net_sales_usd, sales minus returns | Constant currency (static FX) |
| Gross Margin % | (Net Sales minus COGS) / Net Sales        | Returns carry negative COGS   |

| KPI                  | Definition                                                                                 | Rule / edge case                                                |
|----------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Transactions         | Distinct transaction_number with positive net value                                        | Pure return tickets excluded                                    |
| AOV                  | Net Sales / Transactions                                                                   |                                                                 |
| UPT                  | Units sold / Transactions                                                                  | Sale lines only                                                 |
| Average Unit Retail  | Sale net sales / sale units                                                                | Price and mix signal                                            |
| Return Rate          | Returned USD / Gross sale USD                                                              |                                                                 |
| Identified Client    | client_key not equal to -1                                                                 |                                                                 |
| Client Capture Rate  | Identified net sales / total net sales                                                     | Clienteling quality KPI                                         |
| Active Client        | Identified client with net sales above zero in period                                      |                                                                 |
| New Client           | First purchase date falls in period                                                        | Uses is_new_client flag, cross checked with first_purchase_date |
| Retained Client      | Active in period and active in prior 12 months                                             |                                                                 |
| Reactivated Client   | Active in period, not active prior 12 months, purchased before that                        |                                                                 |
| Lapsed Client        | Active prior 12 months, not active in period                                               |                                                                 |
| Tier (recomputed)    | VVIC at or above USD 50k, VIC 15k, Core 3k, Occasional above 0, on calendar year net spend | Calculated dynamically for any selected year                    |
| VIC Concentration    | Share of identified net sales from top 10% clients                                         | 63.1% in 2025 (54 of 538 clients, top 10% rounded up)           |
| Tourist Share        | Tax free net sales / net sales                                                             |                                                                 |
| Selling Advisor      | staff_key where role_type is not Placeholder                                               |                                                                 |
| Attribution Rate     | Net sales on named advisors / store net sales                                              |                                                                 |
| Advisor Productivity | Net sales / active selling advisors (made at least one sale)                               | Annualise by months active                                      |

| KPI              | Definition                                                                                                | Rule / edge case                  |
|------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------|
| Book Coverage    | Assigned clients who purchased in period / assigned clients                                               | Via inactive advisor relationship |
| Orphaned Book    | Clients whose assigned advisor is Resigned                                                                |                                   |
| Sell Through %   | Units sold / (units sold + closing on hand)                                                               | Inventory extension               |
| Weeks of Cover   | Closing on hand units / average weekly units sold last 13 weeks                                           | Inventory extension               |
| Aged Stock %     | On hand cost for SKUs past end_of_life_date or no sale in 180 days / total on hand cost                   | Inventory extension               |
| Entry Conversion | Clients whose entry_division is Fragrance and Beauty or Accessories who later bought Leather Goods or RTW | Luxury funnel KPI                 |

### As built definition notes

- Advisor Client Capture is calculated on sale lines only. A return on an anonymous ticket made one advisor's anonymous sales net negative and pushed the ratio to 107%.
- Cohort Retention = cohort clients active in a year / cohort size. The acquisition year is blanked; those cells read 98 to 99%, not 100%, because first year returns net some clients to zero.
- Weeks of Cover is blank where stock exists but nothing sold in the last 13 weeks. That is overstock, not a stock out, so blanks are left uncoloured.
- Some KPIs ignore the Year slicer by design because they are lifetime or current state: Entry Conversion % (53.9%), Orphaned Clients (70), Contactable % (75.4%).
- Retained + New + Reactivated = 536 against 538 Active Clients in 2025, because New Clients uses the source is\_new\_client flag rather than a derived first purchase rule.
- AOV divides net sales (after returns) by sale tickets only.
- Top 10% Client Share rounds the client count up: 54 of 538 clients hold 63.1% of 2025 identified sales.

## 6. Measure dictionary

All measures live in six measure tables (\_Sales, \_Client, \_Productivity, \_Category, \_Inventory, \_QA), are explicit (implicit measures are discouraged once the calculation group exists), use DIVIDE for ratios and VAR for readability. Every measure below matches the DAX in LinkedIn\_Case\_Study.pbix. Time variants come from the calculation group, not duplicated measures. Mark dim\_date as the date table on dim\_date[date].

### 6.1 Sales core (\_Sales)

```

Net Sales = SUM ( fact_sales[net_sales_usd] )
COGS = SUM ( fact_sales[cogs_usd] )
Gross Margin = [Net Sales] - [COGS]
GM % = DIVIDE ( [Gross Margin], [Net Sales] )

Sale Net Sales = CALCULATE ( [Net Sales], fact_sales[transaction_type] =
"Sale" )
Return Value = - CALCULATE ( [Net Sales], fact_sales[transaction_type] =
"Return" )
Return Rate = DIVIDE ( [Return Value], [Sale Net Sales] )

Transactions =
CALCULATE (
    DISTINCTCOUNT ( fact_sales[transaction_number] ),
    fact_sales[transaction_type] = "Sale"
)
Units Sold = CALCULATE ( SUM ( fact_sales[quantity] ),
fact_sales[transaction_type] = "Sale" )
AOV = DIVIDE ( [Net Sales], [Transactions] )
UPT = DIVIDE ( [Units Sold], [Transactions] )
AUR = DIVIDE ( [Sale Net Sales], [Units Sold] )

Tourist Share = DIVIDE ( CALCULATE ( [Net Sales], fact_sales[is_tax_free] =
TRUE ( ) ), [Net Sales] )
Remote Sale Share = DIVIDE ( CALCULATE ( [Net Sales], fact_sales[channel] =
"Remote Sale" ), [Net Sales] )

```

### 6.2 Time intelligence calculation group

| Item | Expression          |
|------|---------------------|
| CY   | SELECTEDMEASURE ( ) |

| Item  | Expression                                                                                                                                                                                   |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PY    | CALCULATE ( SELECTEDMEASURE (),<br>SAMEPERIODLASTYEAR ( dim_date[date] ) )                                                                                                                   |
| YoY   | VAR cy = SELECTEDMEASURE () VAR py =<br>CALCULATE ( SELECTEDMEASURE (),<br>SAMEPERIODLASTYEAR ( dim_date[date] ) )<br>RETURN IF ( NOT ISBLANK ( cy ) && NOT ISBLANK<br>( py ), cy minus py ) |
| YoY % | Same variables, RETURN DIVIDE ( cy minus py, py );<br>format string expression "0.0%"                                                                                                        |
| YTD   | CALCULATE ( SELECTEDMEASURE (), DATESYTD ( dim_date[date] ) )                                                                                                                                |
| R12M  | CALCULATE ( SELECTEDMEASURE (),<br>DATESINPERIOD ( dim_date[date], MAX ( dim_date[date] ), -12, MONTH ) )                                                                                    |

As built: items carry ordinals 0 to 6 (CY, PY, YoY, YoY %, YTD, PYTD, R12M) so they sort in that order; YoY % has a dynamic format string "0.0%"; for GM % it returns the change in percentage points using ISSELECTEDMEASURE ( [GM %] ). Other items inherit each measure's own format. Test at 2025: CY 3,509,704, PY 4,101,157, YoY minus 591,453, YoY % minus 14.4%.

### 6.3 Price, volume, mix bridge (\_Sales)

Decomposes the change in sale revenue versus prior year at subcategory grain. New subcategories take current AUR as their prior price, so they fall into mix, and the three effects plus the change in returns reconcile exactly to the Net Sales change.

$$\Delta R = \underbrace{\sum_i (P_{1i} - P_{0i}) Q_{1i}}_{\text{Price}} + \underbrace{(Q_1 - Q_0) \bar{P}_0}_{\text{Volume}} + \underbrace{\sum_i P_{0i} Q_{1i} - \bar{P}_0 Q_1}_{\text{Mix}}$$

```
AUR PY = CALCULATE ( [AUR], SAMEPERIODLASTYEAR ( dim_date[date] ) )
Units Sold PY = CALCULATE ( [Units Sold], SAMEPERIODLASTYEAR ( dim_date[date] ) )
```

```
PVM Price =
SUMX (
    VALUES ( dim_product[subcategory] ),
    VAR q1 = [Units Sold]
    VAR p1 = [AUR]
```

```

    VAR p0 = [AUR PY]
    RETURN IF ( NOT ISBLANK ( p0 ) && q1 > 0, ( p1 - p0 ) * q1 )
)

PVM Volume = ( [Units Sold] - [Units Sold PY] ) * [AUR PY]

PVM Mix =
VAR p0bar = [AUR PY]
VAR q1tot = [Units Sold]
VAR p0q1 =
    SUMX (
        VALUES ( dim_product[subcategory] ),
        VAR q1 = [Units Sold]
        RETURN q1 * COALESCE ( [AUR PY], [AUR] )
    )
RETURN p0q1 - p0bar * q1tot

PVM Returns = - ( [Return Value] - CALCULATE ( [Return Value],
SAMEPERIODLASTYEAR ( dim_date[date] ) ) )

```

Test: PVM Price + PVM Volume + PVM Mix + PVM Returns must equal Net Sales YoY for every filter context (QA page).

## 6.4 Client and CRM (\_Client)

```

Identified Net Sales = CALCULATE ( [Net Sales], dim_client[client_key] <> -1
)
Client Capture Rate = DIVIDE ( [Identified Net Sales], [Net Sales] )

Active Clients =
COUNTROWS (
    FILTER ( VALUES ( dim_client[client_key] ), dim_client[client_key] <> -1
&& [Net Sales] > 0 )
)

Prior 12M Active Clients =
VAR _start = MIN ( dim_date[date] )
RETURN
    CALCULATE ( [Active Clients], DATESINPERIOD ( dim_date[date], _start - 1,
-12, MONTH ) )

Retained Clients =
VAR _start = MIN ( dim_date[date] )
VAR _cur = FILTER ( VALUES ( dim_client[client_key] ), dim_client[client_key]
<> -1 && [Net Sales] > 0 )
VAR _prior =

```

```

    CALCULATETABLE (
        FILTER ( VALUES ( dim_client[client_key] ), dim_client[client_key] <>
-1 && [Net Sales] > 0 ),
        DATESINPERIOD ( dim_date[date], _start - 1, -12, MONTH )
    )
RETURN COUNTROWS ( INTERSECT ( _cur, _prior ) )

Retention Rate = DIVIDE ( [Retained Clients], [Prior 12M Active Clients] )

Lapsed Clients = [Prior 12M Active Clients] - [Retained Clients]

Reactivated Clients =
VAR _start = MIN ( dim_date[date] )
VAR _cur = FILTER ( VALUES ( dim_client[client_key] ), dim_client[client_key]
<> -1 && [Net Sales] > 0 )
VAR _prior =
    CALCULATETABLE (
        FILTER ( VALUES ( dim_client[client_key] ), [Net Sales] > 0 ),
        DATESINPERIOD ( dim_date[date], _start - 1, -12, MONTH )
    )
VAR _older =
    CALCULATETABLE (
        FILTER ( VALUES ( dim_client[client_key] ), [Net Sales] > 0 ),
        REMOVEFILTERS ( dim_date ),
        dim_date[date] < EDATE ( _start, -12 )
    )
RETURN COUNTROWS ( EXCEPT ( INTERSECT ( _cur, _older ), _prior ) )

New Clients =
CALCULATE ( DISTINCTCOUNT ( fact_sales[client_key] ),
fact_sales[is_new_client] = TRUE (), dim_client[client_key] <> -1 )

Net Sales per Active Client = DIVIDE ( [Identified Net Sales], [Active
Clients] )

Top 10% Client Share =
VAR _c1 =
    FILTER (
        ADDCOLUMNS ( VALUES ( dim_client[client_key] ), "@s", [Net Sales] ),
        dim_client[client_key] <> -1 && [@s] > 0
    )
VAR _n = ROUNDUP ( COUNTROWS ( _c1 ) * 0.1, 0 )
RETURN DIVIDE ( SUMX ( TOPN ( _n, _c1, [@s], DESC ), [@s] ), SUMX ( _c1, [@s]
) )

```

Dynamic tiering uses a disconnected table Tier Band (Tier, Min USD, Max USD, Sort) with rows VVIC 50,000 to infinity, VIC 15,000 to 50,000, Core 3,000 to 15,000, Occasional 0.01 to 3,000:

```

Clients in Tier =
VAR _min = SELECTEDVALUE ( 'Tier Band'[Min USD] )
VAR _max = SELECTEDVALUE ( 'Tier Band'[Max USD] )
RETURN
    COUNTROWS (
        FILTER (
            VALUES ( dim_client[client_key] ),
            VAR s = [Net Sales]
            RETURN dim_client[client_key] <> -1 && s >= _min && s < _max
        )
    )

Sales in Tier =
VAR _min = SELECTEDVALUE ( 'Tier Band'[Min USD] )
VAR _max = SELECTEDVALUE ( 'Tier Band'[Max USD] )
RETURN
    SUMX (
        FILTER (
            ADDCOLUMNS ( VALUES ( dim_client[client_key] ), "@s", [Net Sales]
        ),
            dim_client[client_key] <> -1 && [@s] >= _min && [@s] < _max
        ),
        [@s]
    )

```

Entry funnel:

```

Entry Converted Clients =
VAR _entry =
    CALCULATETABLE (
        VALUES ( dim_client[client_key] ),
        dim_client[entry_division] IN { "Fragrance and Beauty", "Accessories"
    },
        dim_client[client_key] <> -1
    )
VAR _upgraded =
    CALCULATETABLE (
        VALUES ( fact_sales[client_key] ),
        REMOVEFILTERS ( dim_date ),
        REMOVEFILTERS ( dim_product ),
        dim_product[division] IN { "Leather Goods", "Ready-to-Wear" },
        fact_sales[transaction_type] = "Sale"
    )
RETURN COUNTROWS ( INTERSECT ( _entry, _upgraded ) )

```

Contactability and data completeness (measures replace Power Query flags; messaging opt in uses the phone number):

```

Identified Clients = CALCULATE ( COUNTROWS ( dim_client ),
dim_client[client_key] <> -1 )

Clients with Email =
CALCULATE ( COUNTROWS ( dim_client ), NOT ISBLANK ( dim_client[email] ),
dim_client[client_key] <> -1 )

Clients with Phone =
CALCULATE ( COUNTROWS ( dim_client ), NOT ISBLANK ( dim_client[phone] ),
dim_client[client_key] <> -1 )

Contactable Clients =
COUNTROWS (
    FILTER (
        dim_client,
        dim_client[client_key] <> -1
        && (
            ( dim_client[optin_email] = TRUE () && NOT ISBLANK (
dim_client[email] ) )
            || ( dim_client[optin_phone] = TRUE () && NOT ISBLANK (
dim_client[phone] ) )
            || ( dim_client[optin_messaging] = TRUE () && NOT ISBLANK (
dim_client[phone] ) )
        )
    )
)

```

```

    )
)

Contactable % = DIVIDE ( [Contactable Clients], [Identified Clients] )

```

Expected: 1,507 of 2,000 clients contactable (75.4%); email 1,387, phone 1,011, messaging 1,199.

Client page helpers. Client Status is a disconnected DATATABLE (Retained 1, New 2, Reactivated 3, Lapsed 4) that gives each movement its own bar:

```

Cohort Retention =
VAR _yr = SELECTEDVALUE ( dim_date[year] )
VAR _coh = SELECTEDVALUE ( dim_client[cohort_year] )
RETURN
    IF ( _yr > _coh, DIVIDE ( [Active Clients], COUNTROWS ( dim_client ) ) )

Entry Clients =
CALCULATE (
    COUNTROWS ( dim_client ),
    dim_client[entry_division] IN { "Fragrance and Beauty", "Accessories" }
)

Entry Conversion % = DIVIDE ( [Entry Converted Clients], [Entry Clients] )

Client Movement =
SWITCH (
    SELECTEDVALUE ( 'Client Status'[Status] ),
    "Retained", [Retained Clients],
    "New", [New Clients],
    "Reactivated", [Reactivated Clients],
    "Lapsed", [Lapsed Clients]
)

Client Movement Colour =
SWITCH ( SELECTEDVALUE ( 'Client Status'[Status] ), "Lapsed", "#C0392B",
    "Retained", "#2B2B2B", "#B8962E" )

```

Expected 2025: Retained 249, New 183, Reactivated 104, Lapsed 264; Entry Conversion 53.9% (367 of 681).

## 6.5 Productivity (\_Productivity)

```

Advisor Net Sales = CALCULATE ( [Net Sales], dim_staff[role_type] <>
"Placeholder" )
Attribution Rate = DIVIDE ( [Advisor Net Sales], [Net Sales] )

Active Selling Advisors =
CALCULATE (
    DISTINCTCOUNT ( fact_sales[staff_key] ),
    dim_staff[role_type] <> "Placeholder",
    fact_sales[transaction_type] = "Sale"
)

Advisor Months = SUMX ( VALUES ( dim_date[year_month] ), [Active Selling
Advisors] )
Sales per Advisor Month = DIVIDE ( [Advisor Net Sales], [Advisor Months] )
Advisor Client Capture =
DIVIDE (
    CALCULATE ( [Advisor Net Sales], dim_client[client_key] <> -1,
fact_sales[transaction_type] = "Sale" ),
    CALCULATE ( [Advisor Net Sales], fact_sales[transaction_type] = "Sale" )
)

Advisor Label =
SELECTEDVALUE ( dim_staff[staff_name] ) & " | " & SELECTEDVALUE (
dim_staff[job_title] )

Assigned Clients =
CALCULATE (
    COUNTROWS ( dim_client ),
    TREATAS ( VALUES ( dim_staff[staff_key] ),
dim_client[assigned_advisor_key] )
)

Book Clients Purchased =
VAR _book =
    CALCULATETABLE (
        VALUES ( dim_client[client_key] ),
        TREATAS ( VALUES ( dim_staff[staff_key] ),
dim_client[assigned_advisor_key] )
    )
RETURN CALCULATE ( [Active Clients], REMOVEFILTERS ( dim_staff ), _book )

Book Coverage = DIVIDE ( [Book Clients Purchased], [Assigned Clients] )

Orphaned Clients =

```

```

VAR _resigned =
    CALCULATETABLE ( VALUES ( dim_staff[staff_key] ),
dim_staff[employment_status] = "Resigned" )
RETURN
    CALCULATE ( COUNTROWS ( dim_client ), TREATAS ( _resigned,
dim_client[assigned_advisor_key] ) )

```

## 6.6 Category (\_Category)

```

Mix % of Selection = DIVIDE ( [Net Sales], CALCULATE ( [Net Sales],
ALLSELECTED ( dim_product ) ) )
Iconic Share = DIVIDE ( CALCULATE ( [Net Sales], dim_product[is_iconic] =
TRUE () ), [Net Sales] )
Premium Band Share = DIVIDE ( CALCULATE ( [Net Sales],
dim_product[price_band] = "Premium (5k+)" ), [Net Sales] )
Markdown Share = DIVIDE ( CALCULATE ( [Net Sales], fact_sales[is_markdown] =
TRUE () ), [Net Sales] )
Discount Rate = DIVIDE ( SUM ( fact_sales[discount_local] ) , SUM (
fact_sales[gross_sales_local] ) )

```

Discount Rate is only valid within one currency; show it by market, never summed across markets.

## 6.7 Inventory (\_Inventory, extension table)

```

Closing On Hand Units =
CALCULATE (
    SUM ( fact_inventory_snapshot[on_hand_units] ),
    LASTNONBLANK ( dim_date[date], CALCULATE ( SUM (
fact_inventory_snapshot[on_hand_units] ) ) )
)
Closing On Hand Cost =
CALCULATE (
    SUM ( fact_inventory_snapshot[on_hand_cost_usd] ),
    LASTNONBLANK ( dim_date[date], CALCULATE ( SUM (
fact_inventory_snapshot[on_hand_cost_usd] ) ) )
)
Sell Through % = DIVIDE ( [Units Sold], [Units Sold] + [Closing On Hand
Units] )

Weeks of Cover =
VAR _wk =
    DIVIDE ( CALCULATE ( [Units Sold], DATESINPERIOD ( dim_date[date], MAX (
dim_date[date] ), -91, DAY ) ), 13 )
RETURN DIVIDE ( [Closing On Hand Units], _wk )

```

```

Aged Stock % =
VAR _asof = MAX ( dim_date[date] )
RETURN
    DIVIDE (
        CALCULATE ( [Closing On Hand Cost], KEEPFILTERS ( NOT ISBLANK (
dim_product[end_of_life_date] ) && dim_product[end_of_life_date] < _asof ) ),
        [Closing On Hand Cost]
    )

```

Stock is semi additive: it sums across stores and SKUs but never across dates, which is why LASTNONBLANK replaces SUM over time.

```

Aged Stock Cost =
VAR _asof = MAX ( dim_date[date] )
RETURN
    CALCULATE (
        [Closing On Hand Cost],
        KEEPFILTERS ( NOT ISBLANK ( dim_product[end_of_life_date] ) &&
dim_product[end_of_life_date] < _asof )
    )

```

Expected Dec 2025: 2,476 units, \$486.8K cost, 32.3 weeks of cover, 60.0% sell through, 7.9% aged (\$38.4K).

## 6.8 Targets and QA (\_QA)

```

Sales Target = IF ( ISINSCOPE ( dim_date[date] ), BLANK (), SUM (
fact_targets[sales_target_usd] ) )
Target Attainment % = DIVIDE ( [Net Sales], [Sales Target] )

PVM Check = ( [PVM Price] + [PVM Volume] + [PVM Mix] + [PVM Returns] ) - (
[Net Sales] - CALCULATE ( [Net Sales], SAMEPERIODLASTYEAR ( dim_date[date] )
) )
Tier Mismatch Count = COUNTROWS ( FILTER ( dim_client,
dim_client[client_tier] <> dim_client[Tier 2025 Recomputed] ) )

```

Tier 2025 Recomputed is a calculated column on dim\_client (context transition filters the fact to the row's client):

```

Tier 2025 Recomputed =
VAR s = CALCULATE ( SUM ( fact_sales[net_sales_usd] ), dim_date[year] = 2025
)
RETURN
    SWITCH (
        TRUE (),
        dim_client[client_key] = -1, "Unknown",
        s >= 50000, "VVIC",
        s >= 15000, "VIC",
        s >= 3000, "Core",
        s > 0, "Occasional",
        NOT ISBLANK ( dim_client[first_purchase_date] ), "Lapsed",
        "Prospect"
    )

```

## 6.9 Field parameters and helper measures

Both parameters were created through Modeling > New parameter > Fields (a plain DAX table is not a field parameter). Visuals use the setting Show values of selected field.

```

Metric Selector = {
    ("Net Sales", NAMEOF('_Sales'[Net Sales]), 0),
    ("Units Sold", NAMEOF('_Sales'[Units Sold]), 1),
    ("Transactions", NAMEOF('_Sales'[Transactions]), 2),
    ("Active Clients", NAMEOF('_Client'[Active Clients]), 3),
    ("GM %", NAMEOF('_Sales'[GM %]), 4)
}

Dimension Selector = {
    ("Market", NAMEOF('dim_store'[market_code]), 0),
    ("Store", NAMEOF('dim_store'[store_name]), 1),
    ("Division", NAMEOF('dim_product'[division]), 2),
    ("Category", NAMEOF('dim_product'[category]), 3),
    ("Channel", NAMEOF('fact_sales'[channel]), 4)
}

```

The dynamic chart title reads each label together with its Fields column, because a field parameter label is part of a composite key and SELECTEDVALUE on the label alone fails:

```

Selector Title =
VAR _metric =
    SUMMARIZE ( 'Metric Selector', 'Metric Selector'[Metric Selector],
'Metric Selector'[Metric Selector Fields] )
VAR _dim =
    SUMMARIZE ( 'Dimension Selector', 'Dimension Selector'[Dimension
Selector], 'Dimension Selector'[Dimension Selector Fields] )
VAR _metricName =
    IF ( COUNTROWS ( _metric ) = 1, MAXX ( _metric, 'Metric Selector'[Metric
Selector] ), "Net Sales" )
VAR _dimName =
    IF ( COUNTROWS ( _dim ) = 1, MAXX ( _dim, 'Dimension Selector'[Dimension
Selector] ), "Market" )
RETURN
    _metricName & " by " & _dimName

Net Sales PY = CALCULATE ( [Net Sales], 'Time Intelligence'[Period] = "PY" )

```

Net Sales PY shows how a single calculation item is applied inside a measure, for KPI cards. Units Sold PY and AUR PY (Section 6.3) use SAMEPERIODLASTYEAR directly because the PVM measures need them independent of the calculation group.

## 6.10 Waterfall, headlines, drillthrough and QA

PVM Step is a disconnected DATATABLE (Prior Year 0, Price 1, Volume 2, Mix 3, Returns 4) that drives the waterfall. Helper measures read single calculation group items:

```

Net Sales YoY = CALCULATE ( [Net Sales], 'Time Intelligence'[Period] = "YoY"
)
Net Sales YoY % = CALCULATE ( [Net Sales], 'Time Intelligence'[Period] = "YoY
%" )

PVM Value =
SWITCH ( SELECTEDVALUE ( 'PVM Step'[Step] ),
    "Prior Year", CALCULATE ( [Net Sales], SAMEPERIODLASTYEAR (
dim_date[date] ) ),
    "Price", [PVM Price],
    "Volume", [PVM Volume],
    "Mix", [PVM Mix],
    "Returns", [PVM Returns] )

```

Each story page has one dynamic headline measure (table Headlines) that states the finding and updates with the slicers:

| Measure            | Page                 | Reads at 2025                                                                 |
|--------------------|----------------------|-------------------------------------------------------------------------------|
| Exec Headline      | Executive Summary    | Net sales fell 14.4% vs prior year; Premium band lost \$705K                  |
| Store Headline     | Market and Store     | 11 of 17 stores declined; Lotte Avenuel fell most (-\$179K)                   |
| Category Headline  | Category and Product | Premium share fell from 59.4% to 49.3%; iconic share from 19.7% to 10.3%      |
| Client Headline    | Client and CRM       | Retention fell from 52.0% to 48.5%; 264 clients lapsed                        |
| Advisor Headline   | Advisor Productivity | Named advisors captured 91.9% of sales; 70 clients sit with resigned advisors |
| Inventory Headline | Inventory            | Stock covers 32 weeks of demand; 7.9% of stock value is past end of life      |

```

Exec Headline =
VAR _yoy = [Net Sales YoY %]
VAR _prem = CALCULATE ( [Net Sales YoY], dim_product[price_band] = "Premium (5k+)" )
RETURN
    "Net sales " & IF ( _yoy < 0, "fell ", "grew " ) & FORMAT ( ABS ( _yoy ), "0.0%" )
        & " vs prior year; Premium band " & IF ( _prem < 0, "lost ", "added " )
        & FORMAT ( ABS ( _prem ) / 1e3, "$#,0" ) & "K"

```

The other headlines follow the same pattern (see the PBIP TMDL for full text). Drillthrough header measures use LOOKUPVALUE to follow inactive links without activating them:

```

Store Header =
SELECTEDVALUE ( dim_store[store_name] ) & " | " & SELECTEDVALUE ( dim_store[city] ) & " | "
    & SELECTEDVALUE ( dim_store[store_format] ) & " | " & SELECTEDVALUE ( dim_date[year], "All years" )

Client Header =
SELECTEDVALUE ( dim_client[client_id] ) & " | " & SELECTEDVALUE ( dim_client[Tier 2025 Recomputed] ) & " | "
    & SELECTEDVALUE ( dim_client[nationality] ) & " | Client since "
    & FORMAT ( SELECTEDVALUE ( dim_client[first_purchase_date] ), "mmm yyyy" )

```

```

Assigned Advisor =
VAR _k = SELECTEDVALUE ( dim_client[assigned_advisor_key] )
RETURN
    LOOKUPVALUE ( dim_staff[staff_name], dim_staff[staff_key], _k ) & " ("
        & LOOKUPVALUE ( dim_staff[employment_status], dim_staff[staff_key],
            _k ) & ")"

Preferred Store =
LOOKUPVALUE ( dim_store[store_name], dim_store[store_key], SELECTEDVALUE (
    dim_client[preferred_store_key] ) )

Contact Channel =
SELECTEDVALUE ( dim_client[messaging_app] ) & IF ( [Contactable Clients] = 1,
    " (opted in)", " (not contactable)" )

```

QA framework: QA Tests is a DATATABLE of 11 tests (Test ID, Test, Expected, Tolerance). Every test removes all report filters so results never change with slicers:

```

QA Status =
VAR _a = [QA Actual]
VAR _e = SELECTEDVALUE ( 'QA Tests'[Expected] )
VAR _t = SELECTEDVALUE ( 'QA Tests'[Tolerance] )
RETURN IF ( ABS ( _a - _e ) <= _t, "PASS", "FAIL" )

QA Summary =
VAR _pass = COUNTROWS ( FILTER ( VALUES ( 'QA Tests'[Test ID] ), [QA Status]
    = "PASS" ) )
RETURN _pass & " of " & COUNTROWS ( 'QA Tests' ) & " tests passed"

```

QA Actual is a SWITCH over Test ID; each branch wraps its check in CALCULATE ( ..., REMOVEFILTERS () ) and fixes the year where needed. Section 9 lists the tests.

## 7. Report specification

Seven story pages, two hidden drillthrough pages and a hidden QA page, built on a 1920 x 1080 canvas. The pages read as a diagnosis: what happened, where, why, who, and what to do. Each story page shares one layout: brand box, dynamic headline card (carries the finding), four synced slicers (Year, Market, Channel, Division), a row of six KPI cards, four visuals in a two by two grid, and a footer. Visual titles are descriptive; the headline states the conclusion.

## flowchart LR

A[1 Executive Summary] --&gt; B[2 Market and Store]

A --&gt; C[3 Category and Product]

A --&gt; D[4 Client and CRM]

B --&gt; E[5 Advisor Productivity]

C --&gt; F[6 Inventory]

D --&gt; G[7 Actions]

E --&gt; G

F --&gt; G

B -. drillthrough .-&gt; H[Store 360]

D -. drillthrough .-&gt; I[Client 360]

| Page                   | Headline          | KPI cards                                                                  | Visuals (as built)                                                                                                                                                                                                                         |
|------------------------|-------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Executive Summary    | Exec Headline     | Net Sales, Net Sales YoY %, GM %, Active Clients, AOV, Client Capture Rate | Net Sales Bridge vs Prior Year (waterfall on PVM Step); Monthly Net Sales vs Prior Year (line); Net Sales by Market vs Prior Year (clustered bar); Net Sales Change by Price Band (bar, red below 0, green from 0)                         |
| 2 Market and Store     | Store Headline    | None; the matrix total carries the KPIs                                    | Store Performance vs Prior Year (matrix market to store, arrow icons on YoY %); Net Sales Change Breakdown (decomposition tree); explore chart driven by Metric Selector and Dimension Selector with the Selector Title as a dynamic title |
| 3 Category and Product | Category Headline | None                                                                       | Division Mix by Year (100% stacked column); Net Sales Change by Division and Price Band (matrix with red and green background); Premium and Iconic Share Trend (line, axis from 0); Price vs Volume by Subcategory (scatter, log X axis)   |

| Page                   | Headline           | KPI cards                                                                                                                   | Visuals (as built)                                                                                                                                                                                                                                                                                                            |
|------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 Client and CRM       | Client Headline    | Active Clients, Retention Rate, New Clients, Top 10% Client Share, Entry Conversion %, Contactable %                        | Client Movement vs Prior 12 Months (bar on Client Status, colour by Client Movement Colour); Client Tier Contribution (matrix on Tier Band, data bars); Cohort Retention by Year (heatmap, excludes blank and 2025 cohorts); Top 15 Clients: Spend vs Prior Year (table, Top N 15 by Net Sales PY, excludes ANON)             |
| 5 Advisor Productivity | Advisor Headline   | Active Selling Advisors, Sales per Advisor Month, Attribution Rate, Advisor Client Capture, Book Coverage, Orphaned Clients | Advisor Productivity vs Client Capture (scatter by pos_login, average lines, quadrant labels, Advisor Label tooltip); Sales Attribution by Store (bar, red below 0.9, excludes E-commerce APAC); Advisor Book Coverage (table); Active Orphaned Clients (table filtered to Orphaned Clients = 1)                              |
| 6 Inventory            | Inventory Headline | Closing On Hand Units, Closing On Hand Cost, Weeks of Cover, Sell Through %, Aged Stock %, Aged Stock Cost                  | Weeks of Cover by Store and Division (matrix market to store, diverging gradient centred on 26 weeks, blanks unformatted, excludes E-commerce APAC); Monthly Stock vs Units Sold (combo, secondary axis from 0); Sell Through by Product Line (bar, axis from 0); Aged Stock by Product (table, Discontinued SKUs with stock) |

| Page                | Headline        | KPI cards                                                                                   | Visuals (as built)                                                                                                                                           |
|---------------------|-----------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 Actions           | Static text box | None                                                                                        | Actions table (DATATABLE: #, Action, Evidence, Size of Prize, Owner, Source Page) with a footnote that size of prize assumes partial recovery to 2024 levels |
| Store 360 (hidden)  | Store Header    | Net Sales, Net Sales YoY %, AOV, Active Clients, Attribution Rate, Weeks of Cover           | Monthly line; price band change; Top Advisors (Top N 8, placeholders excluded); Weeks of Cover by Division. Drillthrough on store_name, keep all filters on  |
| Client 360 (hidden) | Client Header   | Net Sales (lifetime), Transactions, AOV, Assigned Advisor, Preferred Store, Contact Channel | Spend by Year; Spend by Division; Purchase History. Drillthrough on client_id, keep all filters off so the whole relationship shows                          |
| QA (hidden)         | QA Summary      | None                                                                                        | Test Results table with PASS and FAIL font colour from QA Colour                                                                                             |

**Interactions:** the Year slicer is set to None on Division Mix by Year, Premium and Iconic Share Trend, and Cohort Retention by Year so they always show all five years.

### Key readings (2025)

| Page                 | Reading                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------|
| Executive Summary    | Price flat, volume +\$0.2M, mix -\$0.8M, returns -\$0.1M: clients kept buying but bought cheaper           |
| Market and Store     | Lotte Avenuel -\$179K, Siam Paragon -\$154K, Sydney Castlereagh -\$115K (-56.7%, down in every price band) |
| Category and Product | Premium RTW -\$386K and Premium Leather Goods -\$319K; Core RTW +\$165K (trade down)                       |
| Client and CRM       | Top 15 clients of 2024 cut spend by a net \$373K, 63% of the decline; 2024 cohort retained only 29%        |

| Page                 | Reading                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Advisor Productivity | Sydney Castlereagh lowest attribution (84%); Hana Smith's output rests largely on one VVIC (\$2.8M lifetime, opted out of all channels) |
| Inventory            | Siam Paragon 12 weeks of cover vs Taipei 101 61 and Breeze Nanshan 71; \$38.4K aged stock                                               |

## Global design rules

1. Theme file maison\_apac\_theme.json: ivory page #FAF8F4, white cards, charcoal #2B2B2B for current year, light grey #C8C8C8 for prior year, gold #B8962E as the single accent, red #C0392B and green #2E7D5B for variance only.
2. Segoe UI Semibold 14 pt titles, 10 to 12 pt body; all tables and matrices use the same font size, horizontal gridlines only, totals off where a total is meaningless.
3. Conditional formatting rules use Number thresholds (for example 0 or 0.9), never Percent, which means position within the range.
4. Field parameter visuals use Show values of selected field; slicers on them are single select.
5. Implicit measures are discouraged by the calculation group, so tooltip, value and size wells take measures (text values wrapped with SELECTEDVALUE).
6. Footer on every page: USD at constant FX. Anonymous sales excluded from client metrics. Synthetic data (inventory page adds that inventory was generated to reconcile with sales).
7. Line smoothing off, axes start at 0 unless a log scale is stated, no subtitles except where they explain a method.

## 8. Security, performance and deployment

The model is built as if for a real regional rollout: market managers see only their market, the file is versioned as code, and refresh is parameterised.

### Row level security

| Role           | Table     | DAX filter                                                                                                             | Notes                                                                              |
|----------------|-----------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Market Manager | dim_store | dim_store[market_code] = LOOKUPVALUE ( 'Security Map'[market_code], 'Security Map'[user_email], USERPRINCIPALNAME () ) | Security Map table: user_email, market_code. Dynamic RLS, one role for all markets |

| Role             | Table      | DAX filter                                                                         | Notes                                                            |
|------------------|------------|------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Regional         | none       | no filter                                                                          | Full access                                                      |
| Advisor PII mask | dim_client | Object level security hides email, phone, first_name, last_name for Market Manager | Clienteling data is sensitive in luxury; shows privacy awareness |

Status: row and object level security are designed but not built in v1. When built, test with View as role for two markets and confirm page 1 totals change while anonymous handling still holds.

## Performance

1. Remove unused columns (advisor\_notes, fx\_to\_usd in fact) to shrink the VertiPaq footprint. Keep email and phone: the Contactable measure reads them.
2. Integer keys only in relationships; no bidirectional filters; no calculated columns on fact\_sales.
3. Run Performance Analyzer on every page; target under 1 second per visual. Capture DAX queries into DAX Studio and check Server Timings for storage engine vs formula engine split on Retained Clients and Top 10% Client Share (the heaviest iterators).
4. Run Best Practice Analyzer (Tabular Editor) and fix all errors and warnings; record the before and after in the README.
5. Limit visuals to 8 or fewer per page; use field parameters instead of duplicate charts.

## Deployment and version control

| Item         | Standard                                                                                                                                                                                                               | Status                                                       |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| File format  | Power BI Project (.pbip) with TMDL so model and report are diffable text in Git                                                                                                                                        | To do at publish                                             |
| Repository   | <a href="https://github.com/jerometanel/apac-retail-analytics">github.com/jerometanel/apac-retail-analytics</a> : /data, /python (profiling, generate_inventory.py), /pbip, /docs (this spec, theme JSON, screenshots) | To do at publish                                             |
| Parameters   | SourcePath and ExtensionPath for the two workbooks; AsOfDate for lifecycle logic                                                                                                                                       | SourcePath and ExtensionPath in progress (O2); AsOfDate done |
| Hidden pages | QA, Store 360, Client 360 hidden from navigation; drillthrough still works                                                                                                                                             | Store 360 and Client 360 done; QA to hide (O10)              |

| Item       | Standard                                                                                                             | Status          |
|------------|----------------------------------------------------------------------------------------------------------------------|-----------------|
| Refresh    | Scheduled refresh via gateway in a real deployment; incremental refresh on fact_sales by transaction_date documented | Documented only |
| Publishing | Publish to web for the public demo (synthetic data only), a short walkthrough video and PDF export                   | To do           |

## 9. Testing and reconciliation

Testing runs in two layers: 11 automated checks on the hidden QA page inside the model, and an independent reconciliation of headline numbers in Python against the raw sheets. Current result: 11 of 11 tests passed.

### In model QA tests (QA page)

| #  | Test                                             | Expected  | Tolerance |
|----|--------------------------------------------------|-----------|-----------|
| 1  | fact_sales row count                             | 14,211    | 0         |
| 2  | Net Sales 2025 (USD)                             | 3,509,704 | 1         |
| 3  | Net Sales 2024 (USD)                             | 4,101,157 | 1         |
| 4  | PVM bridge reconciles to Net Sales change (2025) | 0         | 1         |
| 5  | Stored vs recomputed client tier mismatches      | 14        | 0         |
| 6  | Clients assigned to resigned advisors            | 70        | 0         |
| 7  | Active clients 2025                              | 538       | 0         |
| 8  | Inventory units sold reconcile to fact_sales     | 0         | 0         |
| 9  | Negative stock rows                              | 0         | 0         |
| 10 | Sales rows with no matching client               | 0         | 0         |
| 11 | Target attainment 2025                           | 0.777     | 0.001     |

### Independent reconciliation (Python on the source workbook)

| Check                                      | Expected (2025 unless stated)                               |
|--------------------------------------------|-------------------------------------------------------------|
| Net Sales YoY                              | -14.4% (-\$591,453)                                         |
| Transactions (all tickets) 2025 / 2024     | 1,974 / 1,912                                               |
| AOV 2025 / 2024                            | \$1,778 / \$2,145                                           |
| Return Rate (all years)                    | 3.1%                                                        |
| GM % (all years)                           | 83.0%                                                       |
| Premium band change                        | -\$704,931                                                  |
| Retention Rate 2025 / 2024                 | 48.5% / 52.0%                                               |
| Top 10% Client Share                       | 63.1%                                                       |
| Attribution Rate / Advisor Client Capture  | 91.9% / 88.5%                                               |
| Weeks of Cover / Sell Through / Aged Stock | 32.3 / 60.0% / 7.9%                                         |
| Store 360 Sydney Castlereagh               | \$87.9K, -56.7%, 83.6% attribution, 51 weeks cover          |
| Client 360 C901147                         | \$2.80M lifetime; Hana Smith; Siam Paragon; not contactable |

DAX Transactions counts sale tickets only, so compare it with the SQL count on like for like tickets.

Sample SQL check:

```
SELECT YEAR(transaction_date) AS yr,
       SUM(net_sales_usd) AS net_sales,
       COUNT(DISTINCT transaction_number) AS txns,
       SUM(net_sales_usd) / COUNT(DISTINCT transaction_number) AS aov
FROM fact_sales
GROUP BY YEAR(transaction_date)
ORDER BY yr;
```

## 10. Limitations and roadmap

The data is synthetic and several retail signals are absent, so conclusions are about method, not a real brand.

| Limitation                              | Impact                                                                                  | Mitigation / next version                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Inventory and targets are synthetic     | Stock transfer and target attainment conclusions illustrate method, not a real brand    | Generator is sales consistent and documented; stated in footer, README and case study |
| Size of prize uses planning assumptions | Values such as half recovered are not forecasts                                         | Footnote on the Actions page                                                          |
| Some KPIs are lifetime or current state | Entry Conversion %, Orphaned Clients and Contactable % do not move with the Year slicer | Tooltip or note on the Client and Advisor pages                                       |
| No footfall or traffic                  | Cannot measure conversion rate                                                          | Add synthetic fact_traffic for conversion and staffing ratio                          |
| No staff hours or rosters               | Productivity is per advisor month, not per hour                                         | Add fact_rota for sales per labour hour                                               |
| Static FX                               | No FX impact analysis                                                                   | Add monthly FX table to split constant currency vs reported growth                    |
| Returns not linked to sale              | Return timing approximate; anonymous returns distort ratios                             | Add original_sales_line_id; capture ratios already use sale lines                     |
| SCD2 staff not populated                | Store transfers not tracked                                                             | Load history rows; model already supports it                                          |
| Small sample (2,000 clients)            | VIC metrics volatile                                                                    | Show counts beside percentages                                                        |
| Security not built                      | No market level access control in v1                                                    | Build dynamic RLS and PII masking as designed in Section 8                            |

**Roadmap:** v1 (done) core model, inventory and targets extensions, calculation group, field parameters, 7 story pages, 2 drillthrough pages and QA page with 11 passing tests. v1.1 close open items O2, O5, O6, O8, O9, O10 and publish as PBIP on GitHub. v1.2 dynamic RLS and PII masking, Best Practice Analyzer clean. v2 lapse risk score (RFM features) on Client 360.